

CAMPUS NETWORK INFRASTRUCTURE IMPLEMENTATION

Site-to-Site IPsec VPN, VLAN Segmentation, Firewall Security & UniFi Wireless

Technical Implementation and Evidence Report

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Technologies: Netgate SG-2100 pfSense

Cisco Catalyst 3850

UniFi UAP-AC-Pro

Ubuntu Server

Hyper-V

IKEv2/IPsec

VLAN

DHCP

Firewall

Wireless Networking

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1. Executive Summary

This project documents the design, implementation and verification of a multi-site network infrastructure connecting a DataCentre and Campus environment.

The implementation uses Netgate SG-2100 firewalls to provide secure site-to-site connectivity using IKEv2/IPsec. The Campus network is segmented into separate VLANs for infrastructure management, staff, contractors and guest users.

A Cisco Catalyst 3850 provides VLAN switching and trunk connectivity between the Campus firewall and the UniFi UAP-AC-Pro. The Campus SG-2100 provides Layer 3 gateway services, DHCP and firewall-based traffic control.

A self-hosted UniFi Network Application was deployed on an Ubuntu Server virtual machine in the DataCentre. The UniFi application provides centralised management of the Campus wireless access point through the existing site-to-site VPN.

The completed implementation was verified through VPN status checks, DHCP testing, VLAN connectivity testing, firewall policy testing, wireless connectivity testing and UniFi device adoption.

2. Project Objectives

1. Establish secure communication between the DataCentre and Campus using IKEv2/IPsec.
2. Implement VLAN segmentation on the Campus network.
3. Provide dedicated networks for management, staff, contractors and guests.
4. Configure DHCP services for each VLAN.
5. Implement firewall policies controlling inter-network access.
6. Deploy a self-hosted UniFi Network Application.
7. Adopt and centrally manage the Campus UAP-AC-Pro.
8. Configure separate wireless SSIDs.
9. Verify network connectivity and security policies through testing.
10. Document the complete implementation as technical evidence.

3. Project Scope

The scope of this implementation includes the DataCentre and Campus firewall configuration, site-to-site IPsec VPN, Campus VLANs, Cisco Catalyst 3850 switching, DHCP, firewall segmentation, UniFi Network Application deployment, UAP-AC-Pro adoption and wireless SSID configuration.

4. Network Architecture

4.1 Overall Architecture

The network consists of two logical locations: a DataCentre and a Campus. The two locations are connected using an IKEv2/IPsec site-to-site VPN.

4.2 VLAN and IP Addressing

VLAN	Name	Subnet	Gateway	Purpose
100	Campus-DevMgnt	172.20.0.0/24	172.20.0.1	Infrastructure management
110	Campus-Staff	172.20.1.0/24	172.20.1.1	Staff users
120	Campus-Contractors	172.20.2.0/24	172.20.2.1	Contractors
130	Campus-Guest	172.20.3.0/24	172.20.3.1	Guest users
-	DataCentre	172.20.96.0/24		DataCentre services

5. topology diagram

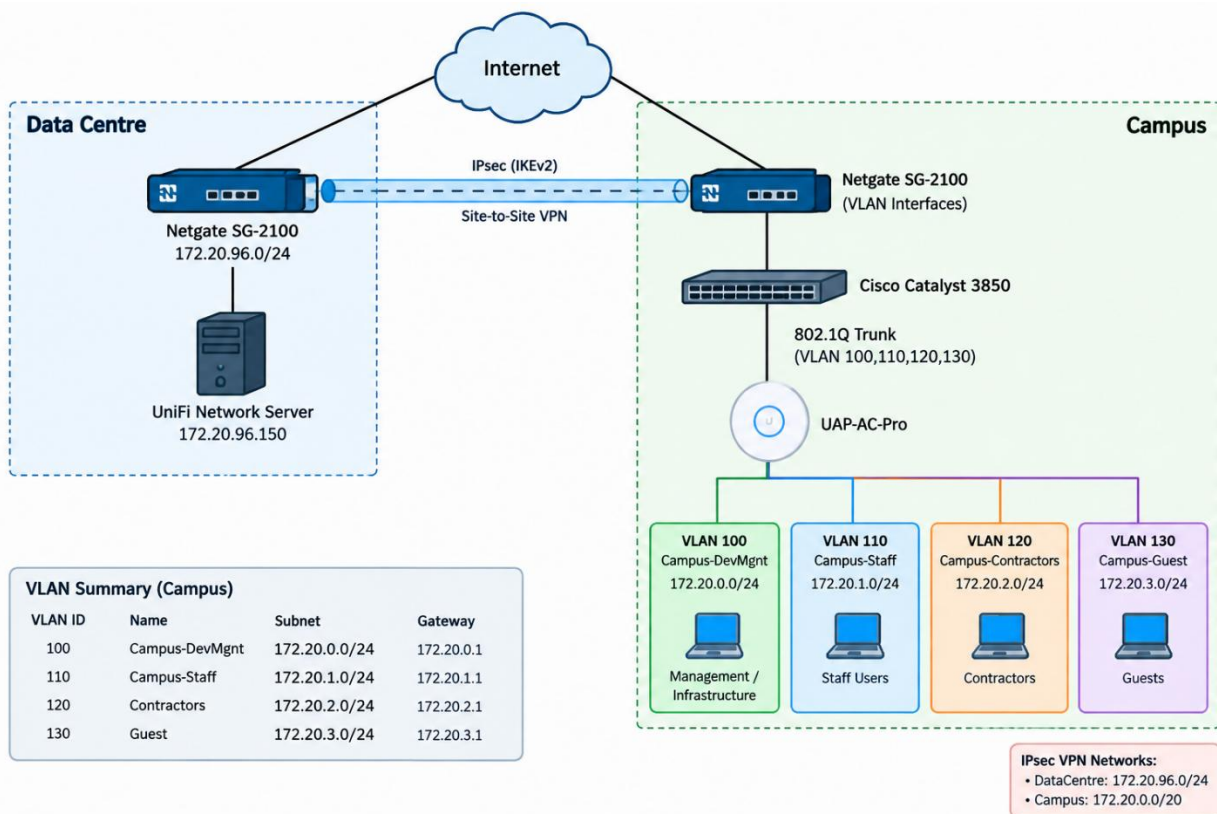


Figure1- Overall Campus/DataCentre Network Topology

6. Site-to-Site IPsec VPN

6.1 DataCentre Phase 1

Status / IPsec / Overview

Overview

Leases

SADs

SPDs

IPsec Status							
ID	Description	Local	Remote	Role	Timers	Algo	Status
con1 #2	S2S to Campus	ID: 192.168.54.116 Host: 192.168.54.116:500 SPI: 21d53b331a6c9459	ID: 192.168.54.100 Host: 192.168.54.100:500 SPI: 7c4a060e42328cc6	IKEv2 Responder	Rekey: 25823s (07:10:23) Reauth: Disabled	AES_CBC (256) HMAC_SHA2_256_128 PRF_HMAC_SHA2_256 MODP_2048	Established 27 seconds (00:00:27) ago Disconnect P1

Figure 2 – DataCentre SG-2100 Phase 1 configuration

The DataCentre firewall was configured as an IKEv2 VPN peer using IPv4 over the WAN interface. Mutual pre-shared key authentication was configured with the local and peer identifiers based on IP addresses.

6.2 DataCentre Phase 2

ID	Description	Local	SPI(s)	Remote	Times	Algo	Stats
con1: #3	DataCentre-to-Campus	172.20.96.0/24	Local: ca69b3ed Remote: cf4ad8e3	172.20.0.0/20	Rekey: 3046s (00:50:46) Life: 3573s (00:59:33) Install: 27s (00:00:27)	AES_CBC (256) HMAC_SHA2_256_128 IPComp: None	Bytes-In: 0 (0 B) Packets-In: 0 Bytes-Out: 0 (0 B) Packets-Out: 0 Installed Disconnect P2

Figure 3 – DataCentre Phase 2 configuration

6.3 Campus Phase 1

pfSense +

System Interfaces Firewall Services VPN Status Diagnostics Help

VPN / IPsec / Tunnels

Tunnels

Mobile Clients

Pre-Shared Keys

Mobile Group Pools

Advanced Settings

IPsec Tunnels									
	ID	IKE	Remote Gateway	Auth/Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
Disable	1	V2	WAN 192.168.54.116	Mutual PSK -	AES (256 bits)	SHA256	14 (2048 bit)	S2S to DataCentre	

Show Phase 2 Entries (1)

Add P1

Delete P1s

6.4 Campus phase 2

	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions
☐ Disable	1	tunnel	172.20.0.0/20	172.20.96.0/24	ESP	AES (256 bits), AES128-GCM (128 bits)	SHA256	Campus-to-DataCentre	Edit Clone Delete
+ Add P2									

[+ Add P1](#) [Delete P1s](#)

6.5 VPN Verification

Figure 4– Established IPsec tunnel

Status / IPsec / Overview

Overview

Leases

SADs

SPDs

IPsec Status

ID	Description	Local	Remote	Role	Timers	Algo	Status
con1 #2	S2S to Campus	ID: 192.168.54.116 Host: 192.168.54.116:500 SPI: 21d53b331a6c9459	ID: 192.168.54.100 Host: 192.168.54.100:500 SPI: 7c4a060e42328cc6	IKEv2 Responder	Rekey: 25823s (07:10:23) Reauth: Disabled	AES_CBC (256) HMAC_SHA2_256_128 PRF_HMAC_SHA2_256 MODP_2048	Established 27 seconds (00:00:27) ago
							Disconnect P1
ID	Description	Local	SPI(s)	Remote	Times	Algo	Stats
con1: #3	DataCentre-to-Campus	172.20.96.0/24	Local: ca69b3ed Remote: cf4ad8e3	172.20.0.0/20	Rekey: 3046s (00:50:46) Life: 3573s (00:59:33) Install: 27s (00:00:27)	AES_CBC (256) HMAC_SHA2_256_128 IPComp: None	Bytes-In: 0 (0 B) Packets-In: 0 Bytes-Out: 0 (0 B) Packets-Out: 0
							Installed Disconnect P2

Figure 5 DataCentre established IPsec tunnel

sense+

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

1

Status / IPsec / Overview

Overview

Leases

SADs

SPDs

IPsec Status

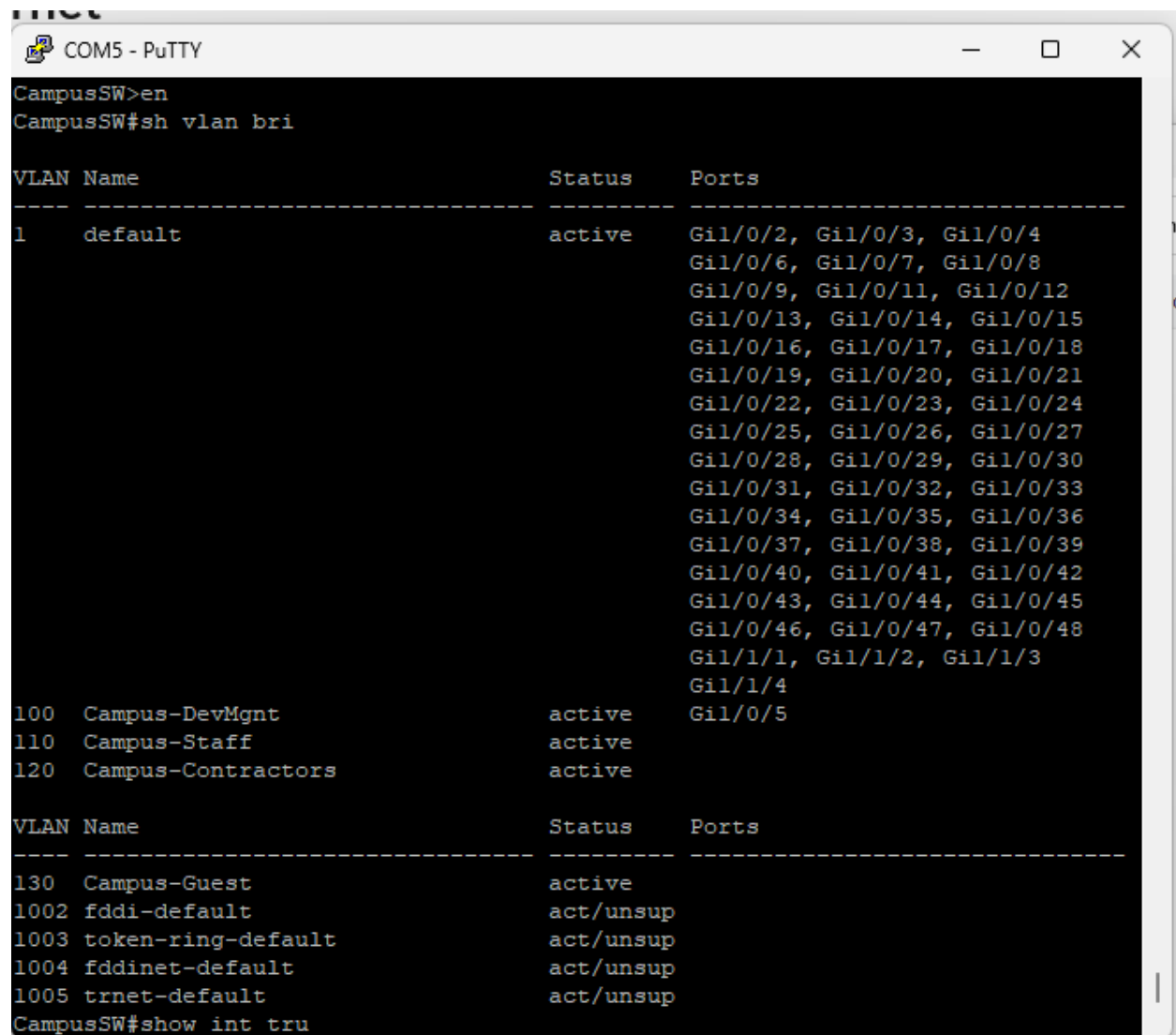
ID	Description	Local	Remote	Role	Timers	Algo	Status
con1 #1	S2S to DataCentre	ID: 192.168.54.100 Host: 192.168.54.100:500 SPI: 7c4a060e42328cc6	ID: 192.168.54.116 Host: 192.168.54.116:500 SPI: 21d53b331a6c9459	IKEv2 Initiator	Rekey: 25550s (07:05:50) Reauth: Disabled	AES_CBC (256) HMAC_SHA2_256_128 PRF_HMAC_SHA2_256 MODP_2048	Established 345 seconds (00:05:45) ago
							Disconnect P1
ID	Description	Local	SPI(s)	Remote	Times	Algo	Stats
con1: #4	Campus-to-DataCentre	172.20.0.0/20	Local: cf4ad8e3 Remote: ca69b3ed	172.20.96.0/24	Rekey: 2883s (00:48:03) Life: 3255s (00:54:15) Install: 345s (00:05:45)	AES_CBC (256) HMAC_SHA2_256_128 IPComp: None	Bytes-In: 0 (0 B) Packets-In: 0 Bytes-Out: 0 (0 B) Packets-Out: 0
							Installed Disconnect P2

Figure 6 campus established IPsec tunnel

7. Cisco Catalyst 3850 Configuration

The Cisco Catalyst 3850 provides Layer 2 switching and 802.1Q trunk connectivity between the Campus firewall and the UniFi wireless access point.

7.1 VLAN Configuration



```
COM5 - PuTTY
CampusSW>en
CampusSW#sh vlan bri

VLAN  Name                               Status  Ports
-----
1     default                               active  Gil/0/2, Gil/0/3, Gil/0/4
                                           Gil/0/6, Gil/0/7, Gil/0/8
                                           Gil/0/9, Gil/0/11, Gil/0/12
                                           Gil/0/13, Gil/0/14, Gil/0/15
                                           Gil/0/16, Gil/0/17, Gil/0/18
                                           Gil/0/19, Gil/0/20, Gil/0/21
                                           Gil/0/22, Gil/0/23, Gil/0/24
                                           Gil/0/25, Gil/0/26, Gil/0/27
                                           Gil/0/28, Gil/0/29, Gil/0/30
                                           Gil/0/31, Gil/0/32, Gil/0/33
                                           Gil/0/34, Gil/0/35, Gil/0/36
                                           Gil/0/37, Gil/0/38, Gil/0/39
                                           Gil/0/40, Gil/0/41, Gil/0/42
                                           Gil/0/43, Gil/0/44, Gil/0/45
                                           Gil/0/46, Gil/0/47, Gil/0/48
                                           Gil/1/1, Gil/1/2, Gil/1/3
                                           Gil/1/4
100    Campus-DevMgmt                         active  Gil/0/5
110    Campus-Staff                          active
120    Campus-Contractors                     active

VLAN  Name                               Status  Ports
-----
130    Campus-Guest                           active
1002   fddi-default                           act/unsup
1003   token-ring-default                     act/unsup
1004   fddinet-default                       act/unsup
1005   trnet-default                         act/unsup
CampusSW#show int tru
```

Figure 7 – VLAN Configuration

7.2 Firewall Uplink

```

-----
130  Campus-Guest          active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default         act/unsup
CampusSW#show int tru

Port      Mode      Encapsulation  Status      Native vlan
-----
Gil/0/1   on        802.1q         trunking    1
Gil/0/10  on        802.1q         trunking    100

Port      Vlans allowed on trunk
-----
Gil/0/1   100,110,120,130
Gil/0/10  100,110,120,130

Port      Vlans allowed and active in management domain
-----
Gil/0/1   100,110,120,130
Gil/0/10  100,110,120,130

Port      Vlans in spanning tree forwarding state and not pruned
-----
Gil/0/1   100,110,120,130
Gil/0/10  100,110,120,130
CampusSW#

```

Figure 8 – Firewall uplink and trunk

The switch uplink to the Campus SG-2100 was configured as an 802.1Q trunk to carry the required VLANs between the firewall and switching infrastructure.

7.3 AP Trunk

```

-----
130  Campus-Guest          active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default         act/unsup
CampusSW#show int tru

Port      Mode      Encapsulation  Status      Native vlan
-----
Gil/0/1   on        802.1q         trunking    1
Gil/0/10  on        802.1q         trunking    100

Port      Vlans allowed on trunk
-----
Gil/0/1   100,110,120,130
Gil/0/10  100,110,120,130

Port      Vlans allowed and active in management domain
-----
Gil/0/1   100,110,120,130
Gil/0/10  100,110,120,130

Port      Vlans in spanning tree forwarding state and not pruned
-----
Gil/0/1   100,110,120,130
Gil/0/10  100,110,120,130
CampusSW#

```

8. Campus Netgate SG-2100 Configuration

8.1 VLAN Interfaces

The screenshot displays the pfSense web interface. At the top is a navigation bar with the pfSense logo and a dropdown menu containing System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. Below the navigation bar is a breadcrumb trail: Interfaces / VLANs. To the right of the breadcrumb are icons for filters, lists, and help. A horizontal menu below the breadcrumb includes tabs for Interface Assignments, Interface Groups, Wireless, VLANs (which is selected and underlined), QinQs, PPPs, GREs, GIFs, VXLANs, Bridges, and LAGGs. The main content area is titled 'VLAN Interfaces' and contains a table with the following data:

Interface	VLAN tag	Priority	Description	Actions
mvneta1 (lan)	100		Campus-DevMgt	
mvneta1 (lan)	110		Campus-Staff	
mvneta1 (lan)	120		Campus-Contractors	
mvneta1 (lan)	130		Campus-Guest	

Below the table, there is an information icon (i) on the left and a green '+ Add' button on the right.

Figure 9 – VLANs Configuration on Firewall

8.2 Interface Assignments

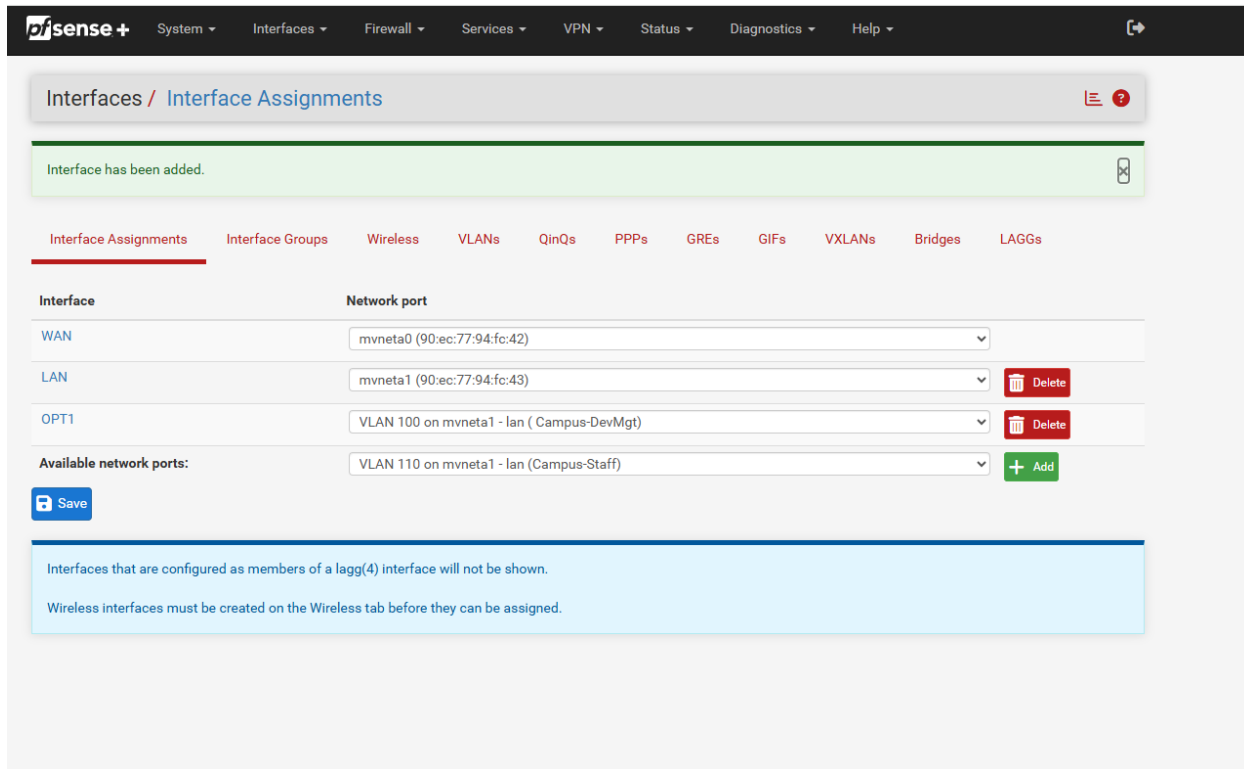
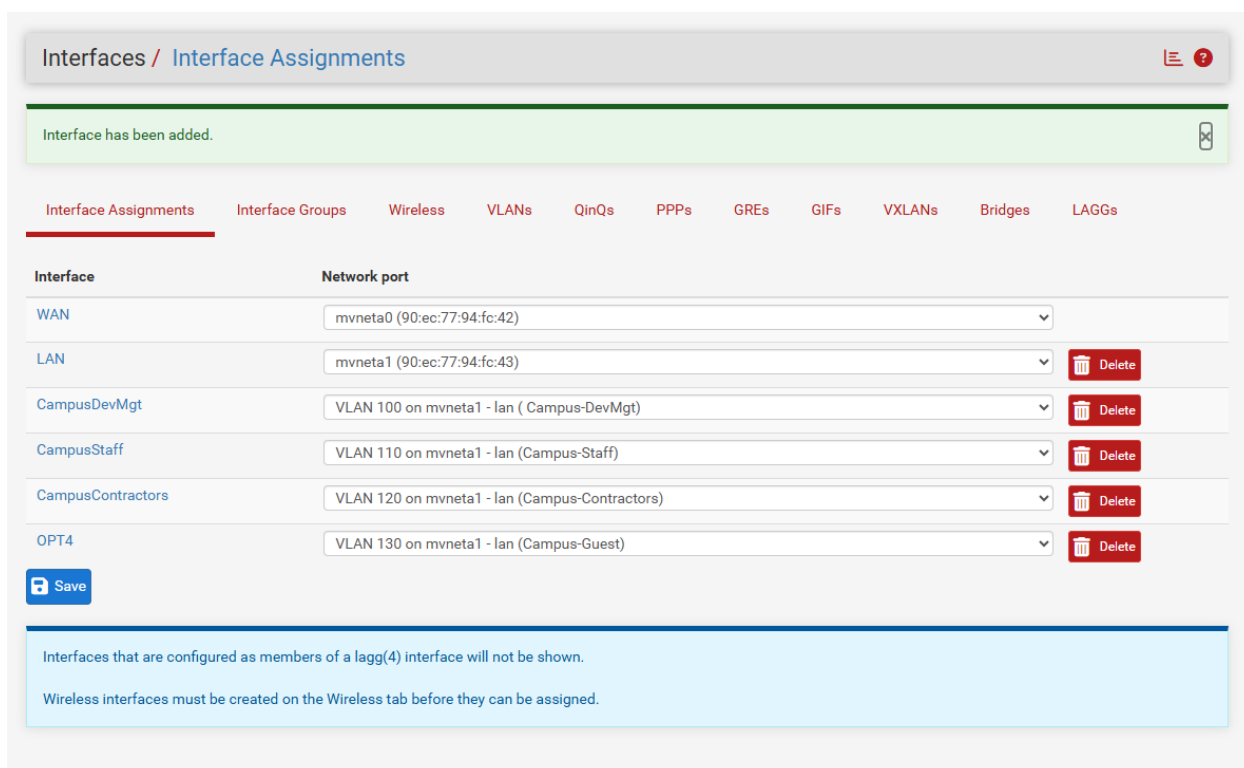


Figure 10 – Interface Assignments



The VLANs were assigned as separate logical interfaces on the Campus SG-2100. Each interface provides the default gateway for its corresponding subnet.

9. DHCP Configuration

DHCP services were configured on the Campus SG-2100 to automatically provide IPv4 addressing, gateway and DNS information to clients on each Campus VLAN.

9.1 Management DHCP

The screenshot shows the pfSense web interface for configuring the DHCP Server on the CAMPUSDEVMGT interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. A notification bar at the top states "The changes have been applied successfully." Below this, a warning message indicates that ISC DHCP has reached end-of-life and will be removed in a future version of Netgate pfSense Plus, suggesting a switch to the DHCP backend. The configuration is for the CAMPUSDEVMGT interface, with other interfaces (LAN, CAMPUSSTAFF, CAMPUSCONTRACTORS, CAMPUSGUEST) also listed. The General Settings section includes options for DHCP Backend (ISC DHCP), Enable (checked), BOOTP (unchecked), Deny Unknown Clients (set to Allow all clients), Ignore Denied Clients (unchecked), and Ignore Client Identifiers (unchecked). The Primary Address Pool section shows the Subnet as 172.20.0.0/24, Subnet Range as 172.20.0.1 - 172.20.0.254, and Address Pool Range as 172.20.0.100 - 172.20.0.200.

Services / DHCP Server / CAMPUSDEVMGT

The changes have been applied successfully.

ISC DHCP has reached end-of-life and will be removed in a future version of Netgate pfSense Plus. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN CAMPUSDEVMGT CAMPUSSTAFF CAMPUSCONTRACTORS CAMPUSGUEST

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on CAMPUSDEVMGT interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	172.20.0.0/24
Subnet Range	172.20.0.1 - 172.20.0.254
Address Pool Range	172.20.0.100 172.20.0.200

9.2 Staff DHCP

pfSense + System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Services / DHCP Server / CAMPUSSTAFF

The changes have been applied successfully.

ISC DHCP has reached end-of-life and will be removed in a future version of Netgate pfSense Plus. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN CAMPUSDEVMTG **CAMPUSSTAFF** CAMPUSCONTRACTORS CAMPUSGUEST

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on CAMPUSSTAFF interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients ▾ When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	172.20.1.0/24
Subnet Range	172.20.1.1 - 172.20.1.254
Address Pool Range	172.20.1.100 172.20.1.200

9.3 Contractor DHCP

System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help

1

Services / DHCP Server / CAMPUSCONTRACTORS

The changes have been applied successfully.

ISC DHCP has reached end-of-life and will be removed in a future version of Netgate pfSense Plus. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN
CAMPUSDEVMGT
CAMPUSSTAFF
CAMPUSCONTRACTORS
CAMPUSGUEST

General Settings

DHCP Backend

ISC DHCP

Enable

☒ Enable DHCP server on CAMPUSCONTRACTORS interface

BOOTP

☐ Ignore BOOTP queries

Deny Unknown Clients

Allow all clients

When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.

Ignore Denied Clients

☐ Ignore denied clients rather than reject
This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.

Ignore Client Identifiers

☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request
This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet

172.20.2.0/24

Subnet Range

172.20.2.1 - 172.20.2.254

Address Pool Range

172.20.2.100

172.20.2.200

9.4 Guest DHCP

pfSense+ System Interfaces Firewall Services VPN Status Diagnostics Help

Services / DHCP Server / CAMPUSGUEST

The changes have been applied successfully.

ISC DHCP has reached end-of-life and will be removed in a future version of Netgate pfSense Plus. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN CAMPUSDEVMT CAMPUSSTAFF CAMPUSCONTRACTORS **CAMPUSGUEST**

General Settings

DHCP Backend ISC DHCP

Enable ☒ Enable DHCP server on CAMPUSGUEST interface

BOOTP ☐ Ignore BOOTP queries

Deny Unknown Clients Allow all clients
 When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed in a static mapping on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.

Ignore Denied Clients ☐ Ignore denied clients rather than reject
 This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.

Ignore Client Identifiers ☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request
 This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet 172.20.3.0/24

Subnet Range 172.20.3.1 - 172.20.3.254

Address Pool Range 172.20.3.100 172.20.3.200

VLAN	DHCP Range	DNS
Management	172.20.0.100–200	172.20.96.10
Staff	172.20.1.100–200	172.20.96.10
Contractors	172.20.2.100–200	172.20.96.10
Guest	172.20.3.100–200	172.20.96.10

9.5 DHCP Verification

pfSense +

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾

1

Status / DHCP Leases

ISC DHCP has reached end-of-life and will be removed in a future version of Netgate pfSense Plus. Visit [System > Advanced > Networking](#) to switch DHCP backend.

Search

Search Term

All ▾

Search

Clear

Enter a search string or *nix regular expression to filter entries.

Leases

	IP Address	MAC Address	Hostname	Description	Start	End	Actions
🕒 ⬆	172.20.0.100	74:ac:b9:66:a2:e3			2026/09/25 04:45:49	2026/09/25 06:45:49	+ +
🕒 ⬆	192.168.1.107	cc:28:aa:84:31:0a	MD-T106-612170		2026/09/25 04:40:16	2026/09/25 06:40:16	+ +
🕒 ⬇	172.20.0.101	c8:f7:50:02:a4:f0	LT-603823		2026/09/25 03:34:41	2026/09/25 05:34:41	+ + ⏻ 🗑

Lease Utilization

Interface	Pool Start	Pool End	Used	Capacity	Utilization
LAN	192.168.1.100	192.168.1.199	1	100	1% of 100
CAMPUSDEVMGT	172.20.0.100	172.20.0.200	2	101	1% of 101

Show All Configured Leases

Clear All DHCP Leases

```
Administrator: Command Pro X + v
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DC01
Primary Dns Suffix . . . . . : wbm.wa
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : wbm.wa

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Hyper-V Network Adapter #2
Physical Address. . . . . : 00-15-5D-01-02-14
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 172.20.96.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 172.20.96.254
DNS Servers . . . . . : 172.20.96.10
NetBIOS over Tcpi . . . . . : Enabled
```

```
DC01 on MD-T106-612168 - Virtual Machine Connection
File Action Media View Help
Select Administrator: Command Prompt
Minimum = 1ms, Maximum = 2ms, Average = 1ms
Control-C
^C
C:\>ping 172.20.0.100 -t

Pinging 172.20.0.100 with 32 bytes of data:
Reply from 172.20.0.100: bytes=32 time=1ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62

Ping statistics for 172.20.0.100:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
Control-C
^C
C:\>ping 172.20.0.101

Pinging 172.20.0.101 with 32 bytes of data:
Reply from 172.20.0.101: bytes=32 time=3ms TTL=126
Reply from 172.20.0.101: bytes=32 time=3ms TTL=126

Ping statistics for 172.20.0.101:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 3ms, Average = 3ms
bytes=32 Control-C
^C
C:\>ping 172.20.0.100

Pinging 172.20.0.100 with 32 bytes of data:
Reply from 172.20.0.100: bytes=32 time=1ms TTL=62
Reply from 172.20.0.100: bytes=32 time=1ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62

Ping statistics for 172.20.0.100:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
Control-C
^C
C:\>_
Microsoft Edge
```

Client-side verification was performed using `ipconfig /all` to confirm that the client received an address from the expected VLAN subnet, together with the appropriate gateway and DNS configuration.

10. Firewall and Security Segmentation

Firewall policies were implemented to control communication between the Campus VLANs and protected DataCentre resources.

10.1 Outbound NAT

Outbound NAT was reviewed to ensure that traffic destined for the remote DataCentre/Campus network is not incorrectly translated before entering the IPsec tunnel.

11. UniFi Network Management Server

A self-hosted UniFi Network Application was deployed on an Ubuntu Server virtual machine hosted in Hyper-V within the DataCentre.

11.1 Ubuntu Server

```
devang@devang:~$ ping 172.20.96.10
PING 172.20.96.10 (172.20.96.10) 56(84) bytes of data.
64 bytes from 172.20.96.10: icmp_seq=1 ttl=128 time=0.391 ms
64 bytes from 172.20.96.10: icmp_seq=2 ttl=128 time=1.03 ms
64 bytes from 172.20.96.10: icmp_seq=3 ttl=128 time=1.23 ms
64 bytes from 172.20.96.10: icmp_seq=4 ttl=128 time=0.827 ms
64 bytes from 172.20.96.10: icmp_seq=5 ttl=128 time=0.664 ms
^C
--- 172.20.96.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4027ms
rtt min/avg/max/mdev = 0.391/0.828/1.225/0.289 ms
devang@devang:~$
```

12. UniFi UAP-AC-Pro

UAP-AC-Pro

↓

Cisco 3850

↓

Campus SG-2100

↓

IPsec VPN

↓

DataCentre

↓

UniFi Server

12. Wireless SSID Configuration

```
Administrator: Command Pro X + v
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 172.20.0.100

Pinging 172.20.0.100 with 32 bytes of data:
Reply from 172.20.0.100: bytes=32 time=1ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62
Reply from 172.20.0.100: bytes=32 time=2ms TTL=62

Ping statistics for 172.20.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Administrator>ping 172.20.0.101

Pinging 172.20.0.101 with 32 bytes of data:
Reply from 172.20.0.101: bytes=32 time=3ms TTL=126
Reply from 172.20.0.101: bytes=32 time=5ms TTL=126
Reply from 172.20.0.101: bytes=32 time=3ms TTL=126
Reply from 172.20.0.101: bytes=32 time=4ms TTL=126

Ping statistics for 172.20.0.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 5ms, Average = 3ms

C:\Users\Administrator>|
```

13. Security Architecture

Network	Access Policy
Management	Infrastructure administration
Staff	Campus/DataCentre/Internet according to policy
Staff	Campus/DataCentre/Internet according to policy
Contractors	Restricted internal access
Guest	Internet access with internal network restrictions

VLAN segmentation reduces the need for all wireless clients to share the same Layer 2 broadcast domain. Firewall policies provide additional Layer 3 access control between the networks.

14. Testing and Verification

Test ID	Test	Expected Result	Result
T01	IPsec Phase 1	Established	Verified
T02	IPsec Phase 2	Established	Verified
T03	Staff DHCP	Correct subnet	verified
T04	Contractor DHCP	Correct subnet	Verified
T05	Guest DHCP	Correct subnet	Verified
T06	Staff → DataCentre	permitted	Verified
T07	Contractor → Staff	Blocked	Verified
T08	Contractor → DataCentre	Blocked	Verified
T09	Guest → DataCentre	Blocked	Verified
T10	Guest → Internet	permitted	Verified
T11	UAP adoption	Successful	verified

15. Final Implementation Summary

The completed implementation provides a segmented Campus network connected securely to the DataCentre through an IKEv2/IPsec VPN. The Campus SG-2100 provides VLAN gateway, DHCP, NAT and firewall services, while the Cisco Catalyst 3850 provides VLAN switching and trunk connectivity.

The UniFi UAP-AC-Pro provides wireless access through separate SSIDs mapped to dedicated VLANs. The UniFi Network Application hosted on Ubuntu Server provides centralised management of the wireless infrastructure.

Security policies were implemented to control communication between Staff, Contractor, Guest and DataCentre networks. Testing was performed to verify VPN operation, DHCP allocation, VLAN connectivity, firewall restrictions and wireless operation.

16. Technical Skills Demonstrated

Networking

IPv4 addressing

Subnetting

VLANs

802.1Q trunking

Layer 3 gateway configuration

DHCP

NAT

Site-to-site VPN

Security

IKEv2/IPsec

AES-256

SHA-256

Firewall segmentation

Network isolation

Guest access restrictions

Wireless security

Infrastructure

Netgate SG-2100

pfSense

Cisco Catalyst 3850

UniFi UAP-AC-Pro

UniFi Network Application

Ubuntu Server

Hyper-V

Troubleshooting

VPN troubleshooting

VLAN troubleshooting

DHCP troubleshooting

Firewall troubleshooting

Wireless troubleshooting

17. Appendix – Evidence

[Appendix A – Additional IPsec Evidence](#)

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Interfaces / OPT1 (mvneta1.100)

General Configuration

Enable

☒ Enable interface

Description

Enter a description (name) for the interface here.

IPv4 Configuration Type

Static IPv4

IPv6 Configuration Type

None

MAC Address

The MAC address of a VLAN interface must be set on its parent interface

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Promiscuous Mode

☐ Enable Promiscuous Mode

Put the interface into permanently promiscuous mode.

Switch port

Select the Switch port to monitor for media state changes

Use the selected Switch port as source for the port state changes.

Static IPv4 Configuration

IPv4 Address

/ 24

IPv4 Upstream gateway

None

+ Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button. On local area network interfaces the upstream gateway should be "none".

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pfSense +

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾

↗

Interfaces / CampusDevMgt (mvneta1.100)

🔍 📄 ?

The changes have been applied successfully.

✕

General Configuration

Enable

☒ Enable interface

Description

Enter a description (name) for the interface here.

IPv4 Configuration Type

Static IPv4 ▾

IPv6 Configuration Type

None ▾

MAC Address

The MAC address of a VLAN interface must be set on its parent interface

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Promiscuous Mode

☐ Enable Promiscuous Mode

Put the interface into permanently promiscuous mode.

Switch port

▾

Use the selected Switch port as source for the port state changes.

Static IPv4 Configuration

IPv4 Address

/ 24 ▾

IPv4 Upstream gateway

▾

+ Add a new gateway

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pfSense +

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Interfaces / CampusStaff (mvneta1.110)

The changes have been applied successfully.

General Configuration

Enable ☒ Enable interface

Description

Enter a description (name) for the interface here.

IPv4 Configuration Type

IPv6 Configuration Type

MAC Address

The MAC address of a VLAN interface must be set on its parent interface

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Promiscuous Mode ☐ Enable Promiscuous Mode

Put the interface into permanently promiscuous mode.

Switch port

Use the selected Switch port as source for the port state changes.

Static IPv4 Configuration

IPv4 Address /

IPv4 Upstream gateway [+ Add a new gateway](#)

pfSense +

System Interfaces Firewall Services VPN Status Diagnostics Help

Interfaces / CampusContractors (mvneta1.120)

The changes have been applied successfully.

General Configuration

Enable ☒ Enable interface

Description CampusContractors

Enter a description (name) for the interface here.

IPv4 Configuration Type Static IPv4

IPv6 Configuration Type None

MAC Address xxxxxxxxxxxx

The MAC address of a VLAN interface must be set on its parent interface

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Promiscuous Mode ☐ Enable Promiscuous Mode

Put the interface into permanently promiscuous mode.

Switch port Select the Switch port to monitor for media state changes

Use the selected Switch port as source for the port state changes.

Static IPv4 Configuration

IPv4 Address 172.20.2.1 / 24

IPv4 Upstream gateway None

+ Add a new gateway

Appendix B – Additional Firewall Evidence

System
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Status / Interfaces

WAN Interface (wan, mvneta0)

Status

up

DHCP

up

Release WAN

Relinquish Lease

MAC Address

90:ec:77:94:fc:42

IPv4 Address

192.168.54.119

Subnet mask IPv4

255.255.255.0

Gateway IPv4

192.168.54.1

IPv6 Link Local

fe80::92ec:77ff:fe94:fc42%mvneta0

DNS servers

192.168.59.131

MTU

1500

Media

1000baseT <full-duplex>

In/out packets

51183/33079 (26.83 MiB/7.39 MiB)

In/out packets (pass)

51183/33079 (26.83 MiB/7.39 MiB)

In/out packets (block)

8175/0 (752 KiB/0 B)

In/out errors

0/0

Collisions

0

LAN Interface (lan, mvneta1)

Status

up

MAC Address

90:ec:77:94:fc:43

IPv4 Address

192.168.1.1

Subnet mask IPv4

255.255.255.0

IPv6 Link Local

fe80::92ec:77ff:fe94:fc43%mvneta1

MTU

1500

Media

2500Base-KX <full-duplex>

In/out packets

22408/37958 (7.40 MiB/28.39 MiB)

In/out packets (pass)

22408/37958 (7.40 MiB/28.39 MiB)

In/out packets (block)

436/0 (52 KiB/0 B)

In/out errors

0/0

Collisions

0

CAMPUSDEVMGT Interface (opt1, mvneta1.100)

Status up 
MAC Address 90:ec:77:94:fc:43
IPv4 Address 172.20.0.1
Subnet mask IPv4 255.255.255.0
IPv6 Link Local fe80::92ec:77ff:fe94:fc43%mvneta1.100
MTU 1500
Media Other <full-duplex>
In/out packets 0/2 (0 B/160 B)
In/out packets (pass) 0/2 (0 B/160 B)
In/out packets (block) 0/2 (0 B/0 B)
In/out errors 0/0
Collisions 0

CAMPUSSTAFF Interface (opt2, mvneta1.110)

Status up 
MAC Address 90:ec:77:94:fc:43
IPv4 Address 172.20.1.1
Subnet mask IPv4 255.255.255.0
IPv6 Link Local fe80::92ec:77ff:fe94:fc43%mvneta1.110
MTU 1500
Media Other <full-duplex>
In/out packets 0/2 (0 B/140 B)
In/out packets (pass) 0/2 (0 B/140 B)
In/out packets (block) 0/3 (0 B/0 B)
In/out errors 0/0
Collisions 0

CAMPUSCONTRACTORS Interface (opt3, mvneta1.120)

Status up 
MAC Address 90:ec:77:94:fc:43
IPv4 Address 172.20.2.1
Subnet mask IPv4 255.255.255.0
IPv6 Link Local fe80::92ec:77ff:fe94:fc43%mvneta1.120
MTU 1500
Media Other <full-duplex>
In/out packets 0/3 (0 B/228 B)
In/out packets (pass) 0/3 (0 B/228 B)
In/out packets (block) 0/2 (0 B/0 B)
In/out errors 0/0
Collisions 0

CAMPUSSTAFF Interface (opt2, mvneta1.110)

Status up 
MAC Address 90:ec:77:94:fc:43
IPv4 Address 172.20.1.1
Subnet mask IPv4 255.255.255.0
IPv6 Link Local fe80::92ec:77ff:fe94:fc43%mvneta1.110
MTU 1500
Media Other <full-duplex>
In/out packets 0/2 (0 B/140 B)
In/out packets (pass) 0/2 (0 B/140 B)
In/out packets (block) 0/3 (0 B/0 B)
In/out errors 0/0
Collisions 0

CAMPUSCONTRACTORS Interface (opt3, mvneta1.120)

Status up 
MAC Address 90:ec:77:94:fc:43
IPv4 Address 172.20.2.1
Subnet mask IPv4 255.255.255.0
IPv6 Link Local fe80::92ec:77ff:fe94:fc43%mvneta1.120
MTU 1500
Media Other <full-duplex>
In/out packets 0/3 (0 B/228 B)
In/out packets (pass) 0/3 (0 B/228 B)
In/out packets (block) 0/2 (0 B/0 B)
In/out errors 0/0
Collisions 0

CAMPUSGUEST Interface (opt4, mvneta1.130)

Status up 
MAC Address 90:ec:77:94:fc:43
IPv4 Address 172.20.3.1
Subnet mask IPv4 255.255.255.0
IPv6 Link Local fe80::92ec:77ff:fe94:fc43%mvneta1.130
MTU 1500
Media Other <full-duplex>
In/out packets 0/3 (0 B/228 B)
In/out packets (pass) 0/3 (0 B/228 B)
In/out packets (block) 0/2 (0 B/0 B)
In/out errors 0/0
Collisions 0

pfSense +

System ▾

Interfaces ▾

Firewall ▾

Services ▾

VPN ▾

Status ▾

Diagnostics ▾

Help ▾



Interfaces / Interface Assignments



Interface AssignmentsInterface GroupsWirelessVLANsQinQsPPPsGREsGIFsVXLANSBridgesLAGGs

Interface	Network port
WAN	mvneta0 (90:ec:77:94:fc:42) ▾
LAN	mvneta1 (90:ec:77:94:fc:43) ▾  Delete
CampusDevMgt	VLAN 100 on mvneta1 - lan (Campus-DevMgt) ▾  Delete
CampusStaff	VLAN 110 on mvneta1 - lan (Campus-Staff) ▾  Delete
CampusContractors	VLAN 120 on mvneta1 - lan (Campus-Contractors) ▾  Delete
CampusGuest	VLAN 130 on mvneta1 - lan (Campus-Guest) ▾  Delete

 Save

Interfaces that are configured as members of a lagg(4) interface will not be shown.

Wireless interfaces must be created on the Wireless tab before they can be assigned.